

**West Bengal State Council of Technical & Vocational
Education and Skill Development**



OBSTACLE AVOIDING LINE FOLLOWER CAR

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2023-2024

CERTIFICATE

This is to certify that this project entitled “**OBSTACLE AVOIDING LINE FOLLOWER CAR** ” by **Debraj Goswami, Rahul Dey, Ganesh Ghosh, Shankha Das, Ranit Khan, Ayan Kumar Das , Subham Haldar** is submitted by partial fulfilment of all the requirements for the degree of DIPLOMA in ELECTRONICS & TELE-COMMUNICATION ENGINEERING from K.G.ENGINEERING INSTITUTE, BISHNUPUR,BANKURA of the WBSCT&VE&SD, KOLKATA, during academic year 2023-2024. This application has been carried out under our supervision.

PROJECT GUIDE

DATE

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ACKNOWLEDGEMENT

First and foremost, we would like to thank our projects guide, Mr. Raja Majhi for his constant support and guidance, without which we would not have been able to attempt this project. His contribution in stimulating suggestions and encouragement, helped us to coordinate the project.

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DECLARATION

We the undersigned solemnly declare that the project “**OBSTACLE AVOIDING LINE FOLLOWER CAR** ” is based on our own work carried out during the course of our study under the supervision of **Mr. Raja Majhi (LECTURER, DEPARTMENT OF ELECTRONICS & TELE-COMMUNICATION ENGINEERING)**.

We further clarify that -

- a) The work in the report is original and has been done by us under the general supervision of our supervisor.
- b) The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other university of India or abroad.
- c) We have followed the guidelines provided by the institute for writing project report.
- d) Whenever we have used materials (data, theoretical analysis, or text) from other sources we have given due credit to them in the text of the report and giving their details in the references.

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Abstract

An intelligent robot called a "line follower" finds and follows a visible line that is embedded in the floor. The predetermined route can be seen as a black line on a white background with sharply contrasted colors. It can recognize the existence of an obstruction and select a different route to its destination. The robot that detects a line and tries to move in the direction of the target can be programmed to follow instructions. Different sensors can be used to detect the line. The robot can identify the path it needs to take using infrared sensors. The ultrasonic sensor is a device that can detect the obstacle by measuring the distance using sound waves and avoiding collision.

The goal of this project is to create a line-following robot with the ability to avoid obstacles. The robot will be equipped with an infrared sensor array to detect the black line on the ground, and an ultrasonic sensor to detect obstacles. A microcontroller will be used to process the sensor data and generate control signals for the robot's motors. The robot's behavior will be determined by a set of algorithms that combine line following and obstacle avoidance techniques. This research will demonstrate the feasibility of designing an autonomous robot that can navigate through a track with a black line while avoiding obstacles.

Keywords : Arduino, IR Sensor, Motor Driver, Servo Motor, Ultrasonic Sensor.

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INTRODUCTION

A line follower robot's main goal is to go following a clearly designated or visibly different line on the ground or floor. Physical lines, like a black line on a white surface or the opposite, can represent this line. To identify the line and make the appropriate modifications to its movement to stay on course, the robot needs sensors normally mounted on the front of the robot.

In addition to the line following, obstacle avoidance is another crucial feature incorporated into the robot. It allows the robot to detect obstacles or objects in its path and take appropriate actions to avoid collisions. Obstacle detection can be achieved using various sensors, such as ultrasonic and infrared sensors. The combination of line following and obstacle avoidance enables the robot to navigate through complex paths, follow a line with curves and intersections, and avoid obstacles along the way.

To implement a line follower with obstacle avoiding capabilities, the robot typically consists of a programmable board as its brain, sensors for line detection and obstacle detection, motors or wheels for movement, and drive the motors accordingly. Building a line follower with an obstacle-avoiding robot requires a combination of hardware and software skills, including electronics, programming, and sensor integration. It offers a practical and interactive way to learn about robotics, sensor technology, and autonomous navigation systems.

1.1 Purpose

- The purpose of this project is to build a robot that can autonomously follow a black line on the ground while avoiding obstacles.
- It combines line-following capabilities with obstacle detection and avoidance.

1.2 Scope

- The scope includes designing and implementing the hardware and software components necessary for the robot.
- It covers both the line-following behavior and the obstacle-avoidance feature.

1.3 Goal & Objectives

- Goal 1: Develop a functional line follower robot that can accurately follow a predefined path.
- Goal 2: Enhance the robot to detect obstacles (using an ultrasonic sensor) and adjust its path to avoid collisions.

- Objective 1: Assemble the necessary components (Arduino, motors, sensors, etc.).
- Objective 2: Write code to control the motors based on sensor inputs.
- Objective 3: Test and fine-tune the robot's behaviour.

1.4 Definitions, acronym & abbreviations

Term	Description
IR	Infrared
L298N	Motor driver module
HC-SR04	Ultrasonic distance sensor
ENA, ENB	Enable pins for motor A and motor B
IN1, IN2, ...	Input pins for motor control
Vin	Voltage input
GND	Ground (common reference)

Literature review

2.1 Arduino Uno

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller.

Key features of the Arduino Uno:

- 14 digital input/output pins: These pins can be used for various tasks, including controlling motors, reading sensors, and interacting with other devices.
- 6 analog inputs: These allow you to read analog signals (such as sensor values) with precision.
- 16 MHz ceramic resonator: Provides accurate timing for the microcontroller.
- USB connection: Allows you to program the board and communicate with your computer.
- Power jack: Used for external power supply.
- ICSP header: For in-circuit programming.
- Reset button: Resets the microcontroller..

2.2 Arduino Language:

- The Arduino programming language is based on C++ with some additional functions and libraries designed specifically for Arduino boards.
- It's a simplified version of C++, making it accessible for both beginners and experienced programmers.
- You don't need an in-depth understanding of C++ to start writing Arduino programs (called "sketches").
- The language includes functions for controlling the board, handling digital and analog I/O, math operations, and more

2.3 Obstacle Avoiding

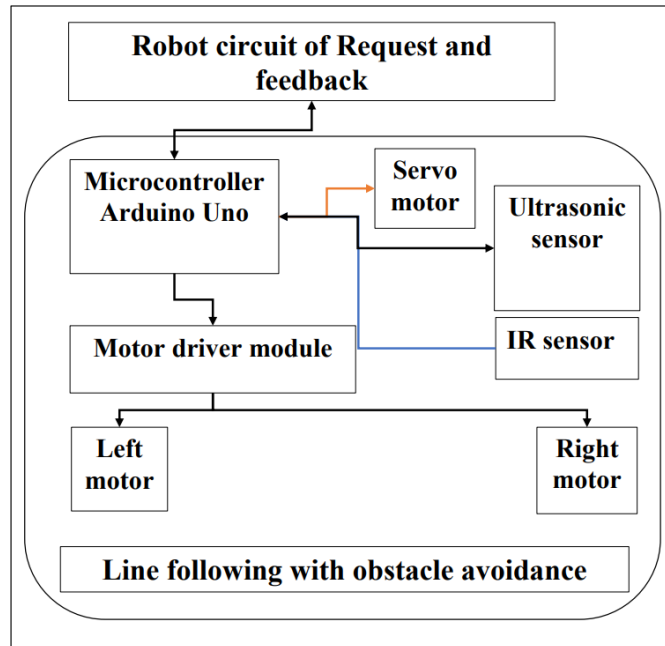
The Arduino Uno serves as the brain of the robot in this project.

It reads input from:

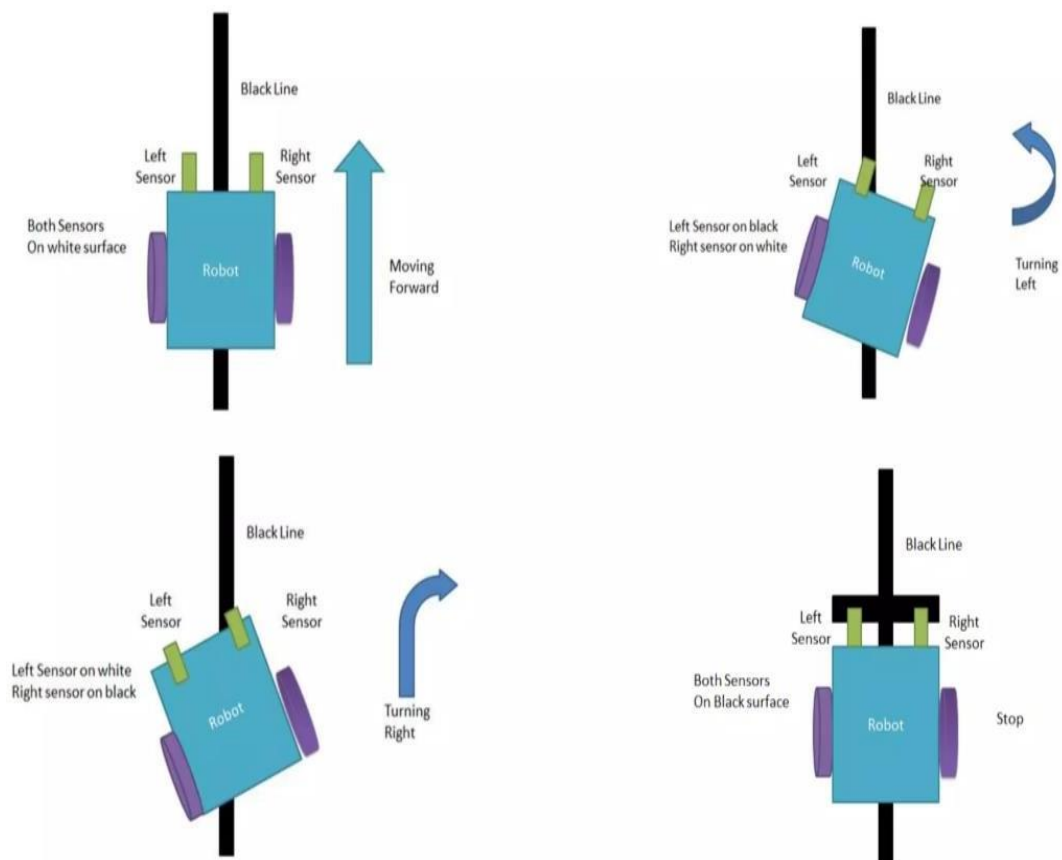
- IR sensors: Used for line following (tracing the black line on the ground).
- Ultrasonic sensor (HC-SR04): Used for obstacle detection and avoidance.
- Based on sensor data, the Arduino controls the motors, adjusting their speed and direction to follow the line while avoiding obstacles.
- The Arduino's versatility, ease of programming, and compatibility with various sensors and motor drivers make it an ideal choice for such projects

Basic Obstacle Avoidance Process

3.1 Working model



3.2 Proposed system



Requirement Specification

Our purpose is to --

- Design an autonomous robot that can follow a black line on the ground while avoiding obstacles.
- Combine line-following capabilities with obstacle detection and avoidance.

4.1 Feasibility Study

The feasibility study investigates the problem and the information needs of the system. It seeks to determine the resources required to provide an information systems solution, the cost and benefits of such a solution, and find the possible problem with their solution. In short, a feasibility analysis evaluates the project's potential for success. Following Feasibility Analysis was performed before working on the project:

4.1.1 Technical feasibility

- Arduino Uno is widely available, affordable, and suitable for this project.
- L298N motor driver provides motor control capabilities.
- IR sensors and ultrasonic sensors are commonly used for line following and obstacle detection.
- Servo motor allows precise steering adjustments..

4.1.2 Operational feasibility

- The robot can operate indoors or on controlled tracks.
- It requires a flat surface with a visible black line.
- Users need basic knowledge of Arduino programming and electronics.

4.1.3 Economical feasibility:

- The components (Arduino, sensors, motor driver) are cost-effective.
- Custom-built chassis can be economical.

4.1.4 Time feasibility

- Building and programming the robot can be completed within a reasonable timeframe.

4.1.5 Safety feasibility:

- Ensure safe voltage levels for components.
- Avoid sharp edges in the design.

4.2 System requirement

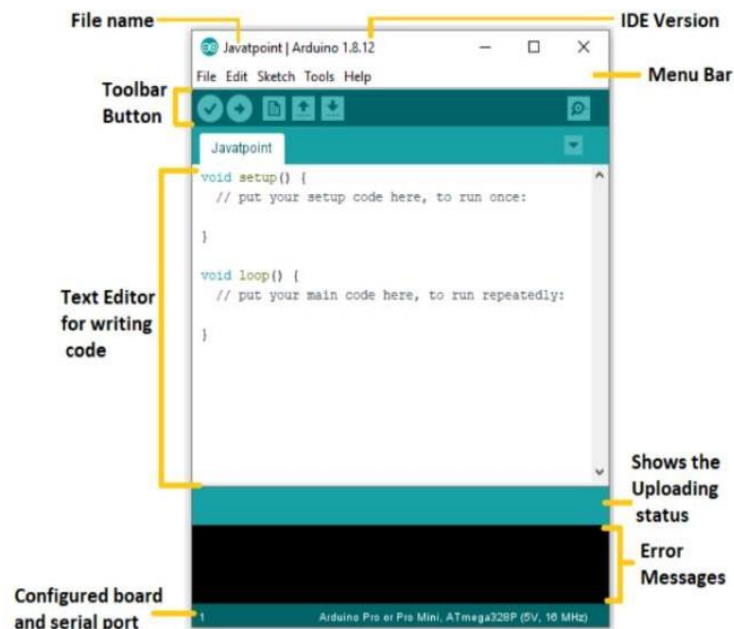
4.2.1 Operating system

- Windows 8 or more

4.2.2 Software requirement

- Arduino IDE

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment. The Arduino IDE will appear as:



4.2.3 Hardware requirement

Any good operatable machine or laptop having those features:

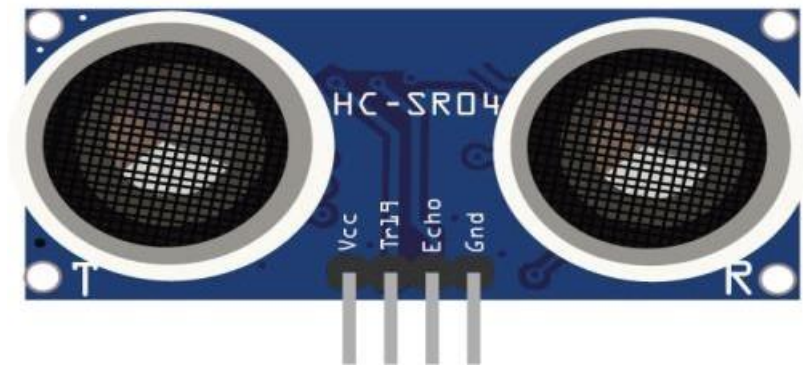
- Processor : any processor above 1GHz
- RAM : 4GB
- Hard Disk : 10GB
- Input Device : mouse and keyboard
- Output device/components : monitor or display
Arduino Uno, Breadboard,

wires, USB connector, two 4.5 volt DC battery , jumper wire , robot car chassis, ultrasonic sensor ,switch, two IR sensor , servo motor , motor driver , 4 wheel

Components

- **HC-SR04(Ultrasonic sensor)**

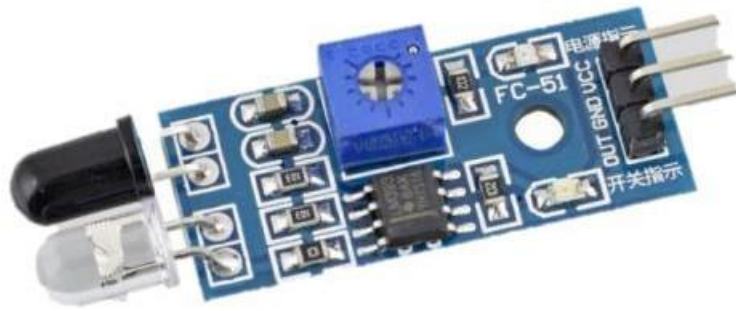
An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. An ultrasonic sensor uses a transducer to send and receive ultrasonic pulses that relay back information about an object's proximity. High-frequency sound waves reflect from boundaries to produce distinct echo patterns. To make a line follower robot with obstacle detection ability, it is attached with an ultrasonic sensor, which is a device that can measure the distance between an object and a robot sensor by using ultrasonic waves.



- **IR-Sensor**

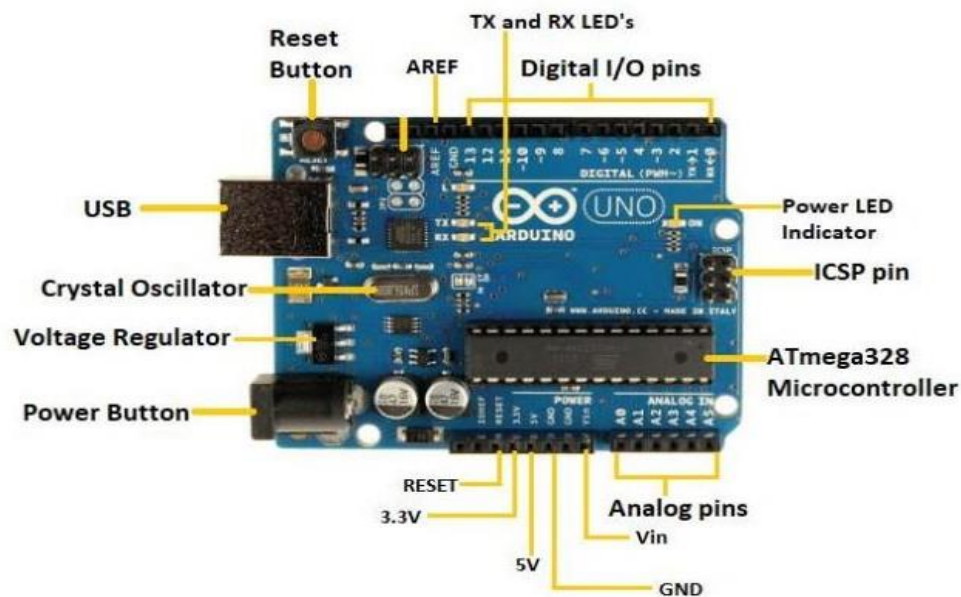
An infrared sensor is an electronic device, that emits in order to sense some aspects of the surroundings. An IR sensor can measure the heat of an object as well as detects the motion. These types of sensors measure only infrared radiation, rather than emitting it that is called a passive IR sensor. Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation. These types of radiations are invisible to our eyes, which can be detected by an infrared sensor. The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode that is sensitive to IR light of the same wavelength as that emitted by the IR LED. When IR light falls on the photodiode, the resistances and the output voltages will

change in proportion to the magnitude of the IR light received.



- **Arduino Uno**

The Arduino UNO is a standard board of Arduino. Here UNO means 'one' in Italian. It was named as UNO to label the first release of Arduino Software. It was also the first USB board released by Arduino. It is considered as the powerful board used in various projects. Arduino.cc developed the Arduino UNO board. Arduino UNO is based on an ATmega328P microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits.



The Arduino UNO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on both online and offline platforms. The IDE is common to all available boards of Arduino.

Let's discuss each component in detail.

- o **ATmega328 Microcontroller**- It is a single chip Microcontroller of the ATmel family. The processor code inside it is of 8-bit. It combines Memory (SRAM,

EEPROM, and Flash), Analog to Digital Converter, SPI serial ports, I/O lines, registers, timer, external and internal interrupts, and oscillator.

- o **ICSP pin** - The In-Circuit Serial Programming pin allows the user to program using the firmware of the Arduino board.

- o **Power LED Indicator**- The ON status of LED shows the power is activated. When the power is OFF, the LED will not light up.
- o **Digital I/O pins**- The digital pins have the value HIGH or LOW. The pins numbered from D0 to D13 are digital pins.

- o **TX and RX LED's**- The successful flow of data is represented by the lighting of these LED's.

- o **AREF**- The Analog Reference (AREF) pin is used to feed a reference voltage to the Arduino UNO board from the external power supply.

- o **Reset button**- It is used to add a Reset button to the connection.

- o **USB**- It allows the board to connect to the computer. It is essential for the programming of the Arduino UNO board.

- o **Crystal Oscillator**- The Crystal oscillator has a frequency of 16MHz, which makes the Arduino UNO a powerful board.

- o **Voltage Regulator**- The voltage regulator converts the input voltage to 5V.

- o **GND**- Ground pins. The ground pin acts as a pin with zero voltage.

- o **Vin**- It is the input voltage.

- o **Analog Pins**- The pins numbered from A0 to A5 are analog pins. The function of Analog pins is to read the analog sensor used in the connection. It can also act as GPIO (General Purpose Input Output) pins.

The technical specifications of the Arduino UNO are listed below:

- o There are 20 Input/Output pins present on the Arduino UNO board. These 20 pins include 6 PWM pins, 6 analog pins, and 8 digital I/O pins.

- o The PWM pins are Pulse Width Modulation capable pins.

- o The crystal oscillator present in Arduino UNO comes with a frequency of 16MHz.

- o It also has a Arduino integrated WiFi module. Such Arduino UNO board is based on the Integrated WiFi ESP8266 Module and ATmega328P microcontroller.

- o The input voltage of the UNO board varies from 7V to 20V.

- o Arduino UNO automatically draws power from the external power supply. It can also draw power from the USB.

- **L298N Motor Driver**

A motor driver is an integrated circuit chip which is usually used to control motors in autonomous robots. Motor driver act as an interface between Arduino and the motors . Here we have used L298N motor driver . The L298N motor driver includes an H-bridge (controls the motor direction and speed), input pins (receive control signals), output pins (connect to the motor), and current sensing/protection circuitry (monitors and protects against overcurrent or short circuits). This motor driver integrated circuit can control two motors at once. The arrangement of motors is known as H-Bridge. An electronic circuit called an "H-Bridge-It" allows a voltage to be applied across a load in either way. It enables complete control of a conventional electric DC motor by a circuit. In other words, an H-bridge enables electronic control of the motor's forward, reverse, brake, and coast functions via a microcontroller, logic chip, or remote control. Figure shows the L298N Motor Driver and its label.

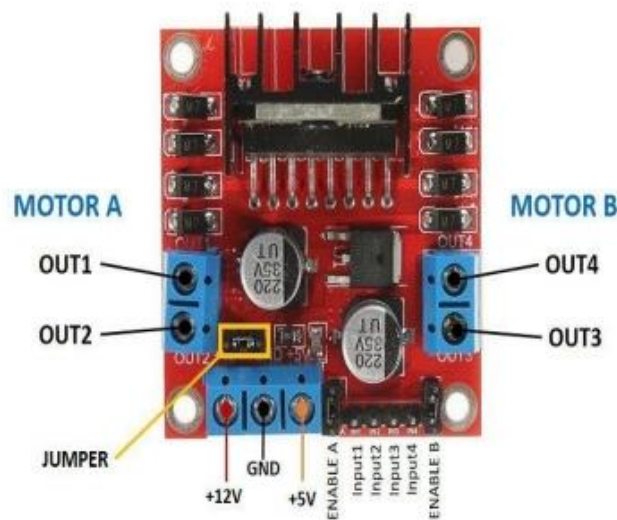


Figure 5: L298 Motor Driver

- **SG90 Servo Motor:**

A servo motor is a type of motor that can rotate with great precision. Normally this type of motor consists of a control circuit that provides feedback on the current position of the motor shaft, this feedback allows the servo motors to rotate with great precision. If you want to rotate an object at some specific angles or distance, then you use a servo motor. It is just made up of a simple motor which runs through a servo mechanism. If motor is powered by a DC power supply then it is called DC servo motor, and if it is AC-powered motor then it is called AC servo motor. For this tutorial, we will be discussing only about the DC servo motor working. The SG90 servo motor comprises a DC motor (generates rotational motion), gear train (amplifies motion), position feedback potentiometer (provides position feedback), and control circuitry (controls motor movement based on input signals).



- **Switch**

In electrical engineering, a switch is an electrical component that can "make" or "break" an electrical circuit, interrupting the current or diverting it from one conductor to another. The mechanism of a switch removes or restores the conducting path in a circuit when it is operated. It may be operated manually, for example, a light switch or a keyboard button, may be operated by a moving object such as a door, or may be operated by some sensing element for pressure, temperature or flow. A switch will have one or more sets of contacts, which may operate simultaneously, sequentially, or alternately. Switches in high-powered circuits must operate rapidly to prevent destructive arcing, and may include special features to assist in rapidly interrupting a heavy current. Multiple forms of actuators are used for operation by hand or to sense position, level, temperature or flow



- **Jumper Wires**

Most, if not all, electronics suppliers stock jumper wire in various lengths and assortments. These wires are commonly used with breadboards and other prototyping tools like Arduino. Jumper wires make changing circuits as simple as possible. A jumper wire may appear uncomplicated, and it doesn't get much more basic than other wires or cables. But there are tiny details you need to pay attention to. Generally, jumpers are tiny metal connectors used to close or open a circuit part. They have two or more connection points, which regulate an electrical circuit board. Their function is to configure the settings for computer peripherals, like the motherboard. Suppose your motherboard supported intrusion detection. A jumper can be set to enable or disable it. Jumper Wires are electrical wires with connector pins at each end. They are used to connect two points in a circuit without soldering. You can use jumper wires to modify a circuit or diagnose problems in a circuit. Further, they are best used to bypass a part of the circuit that does not contain a resistor and is suspected to be bad. This includes a stretch of wire or a switch. Suppose all the fuses are good and the component is not receiving power; find the circuit switch. Then, bypass the switch with the jumper wire.



- **Lithium-Ion Battery**

A lithium-ion (Li-ion) battery is an advanced battery technology that uses lithium ions as a key component of its electrochemistry. During a discharge cycle, lithium atoms in the anode are ionized and separated from their electrons. The lithium ions move from the anode and pass through the electrolyte until they reach the cathode, where they recombine with their electrons and electrically neutralize. The lithium ions are small enough to be able to move through a micro-permeable separator between the anode and cathode.



Two 4.5 V DC battery is used in this project.

Functional Requirements

The requirements that specify what all services a system can provide the end-user are called the functional requirements. These define exactly what functions the system can do. The functional requirements are closely related to the user requirement specifications. This may include calculations, data processing, technical operations and other such functionality that aim to fulfill the application objectives. These are captured in the form of use cases, which are the system responses to events by external agents or internal deadlines.

The technical architecture of the system is determined by these requirements.

Key ID	Title	Requirements	Dependency	Precondition	Post-Condition	Restriction / Risk
FR1	Line Following Behavior	The robot must accurately follow a black line on the ground.	IR sensors (left and right)	The robot is placed on a track with a visible black line.	The robot successfully traces the line.	The line may be imperfect or disrupted due to dirt or wear.
FR2	Obstacle Detection	The robot must detect obstacles using an ultrasonic sensor (HC-SR04).	Ultrasonic sensor (HC-SR04)	The robot is in an environment with obstacles (e.g., walls, objects).	The robot detects obstacles within a specified range.	False positives or false negatives in obstacle detection.
FR3	Obstacle Avoidance	When an obstacle is detected, the robot must adjust its path to avoid collisions.	FR1, FR2	Obstacles are detected by the ultrasonic sensor.	The robot successfully avoids obstacles and continues following the line.	The robot may need to make sudden turns, affecting stability.
FR4	Motor Control	The robot's motors must be controlled based on sensor inputs (line following and obstacle detection).	L298N motor driver module	The robot receives input from IR sensors and ultrasonic sensor.	The robot adjusts motor speeds and directions accordingly.	Incorrect motor control may lead to erratic behavior.
FR5	Power Management	The robot must operate using a 7-12V battery pack.	Battery pack	The battery pack is sufficiently charged.	The robot operates without power interruptions.	Low battery voltage may affect performance.

FR6	Robustness and Durability	The robot should withstand minor shocks and operate reliably.	Sturdy chassis and components	The robot is assembled securely and robustly.	The robot continues functioning even after minor collisions or jolts.	Excessive shocks may damage components.
FR7	Calibration and Tuning	The robot's sensor thresholds and response times must be adjusted for optimal performance.	Sensor calibration process	The robot is stationary and ready for calibration.	The robot's sensors respond accurately to line and obstacle cues.	Incorrect calibration may lead to tracking errors or false alarms.
FR8	User-Friendly Interface	The robot's behavior should be intuitive and easy to understand.	Clear instructions for users	Users can interact with the robot (e.g., start, stop, calibrate).	Users can operate the robot without confusion.	Complexity or lack of user guidance may hinder usability.

Non-functional requirements

The requirements that describe how the system runs are the non-functional requirements. These requirements determine the way the system behaves and the constraints on it. These criteria are used to judge the system in terms of performance, reliability and security.

Some of the non-functional requirements of this project are:

6.1 Performance

- The robot should respond quickly to changes in line direction and obstacle detection.
- Line-following accuracy should be high, minimizing deviations from the desired path.
- Obstacle avoidance should occur promptly to prevent collisions.

6.2 Robustness and Durability

- The robot should withstand minor shocks or bumps without malfunctioning.
- Components (such as sensors, motors, and chassis) should be securely attached to prevent dislodging during operation.

6.3 Power Efficiency

- Optimize power usage to prolong battery life.
- Minimize energy consumption during idle states.

6.4 Scalability

- Design the robot to be easily expandable or modifiable for future enhancements (e.g., additional sensors, improved algorithms).

6.5 Usability and User Interface

- The robot's behavior should be intuitive for users.
- Clear instructions for setup, calibration, and operation should be provided.

6.6 Safety

- The robot should not pose any safety risks to users or surroundings.
- Motors should stop immediately upon emergency shutdown.

6.7 Maintenance and Serviceability

- Components should be accessible for maintenance or replacement.
- Troubleshooting guides should be available for common issues.

System design

7.1 Block diagram

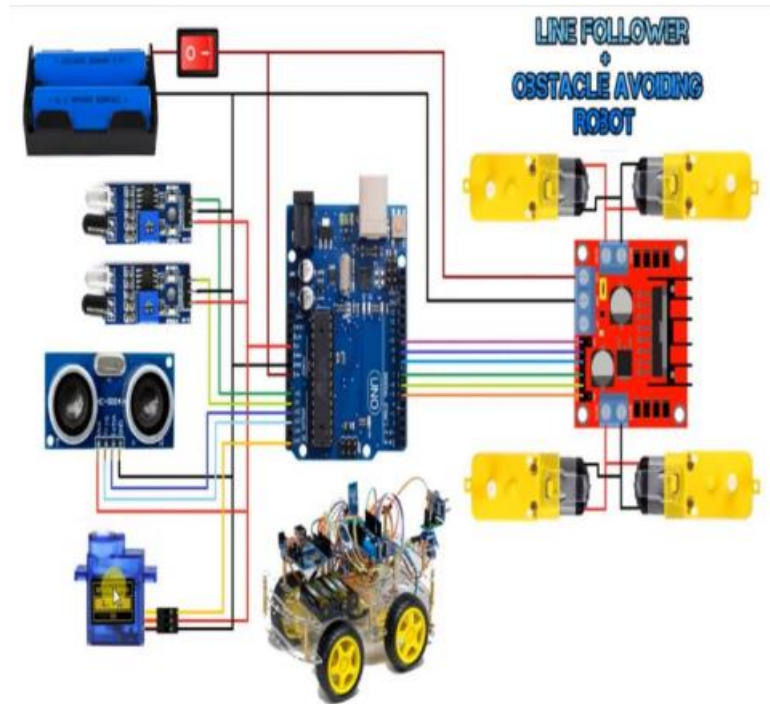


Fig 1: Circuit connection

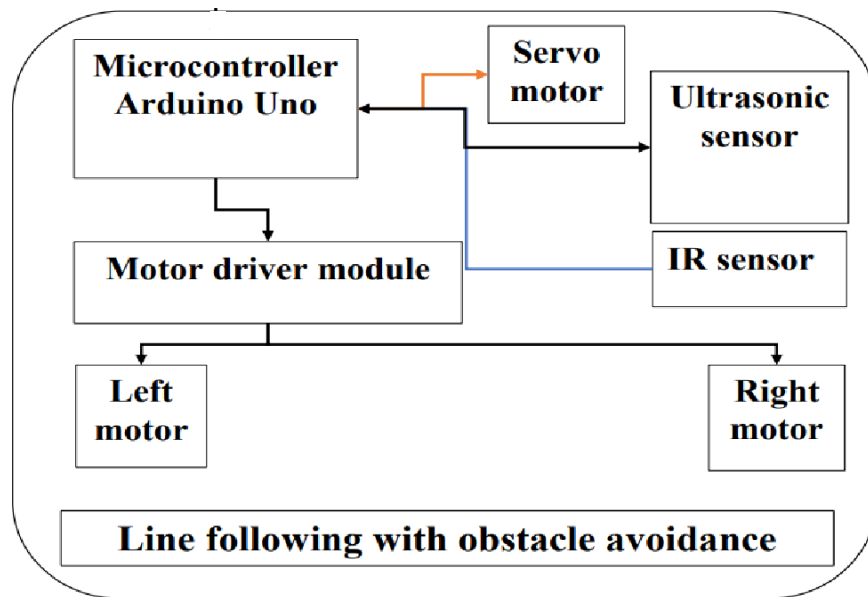
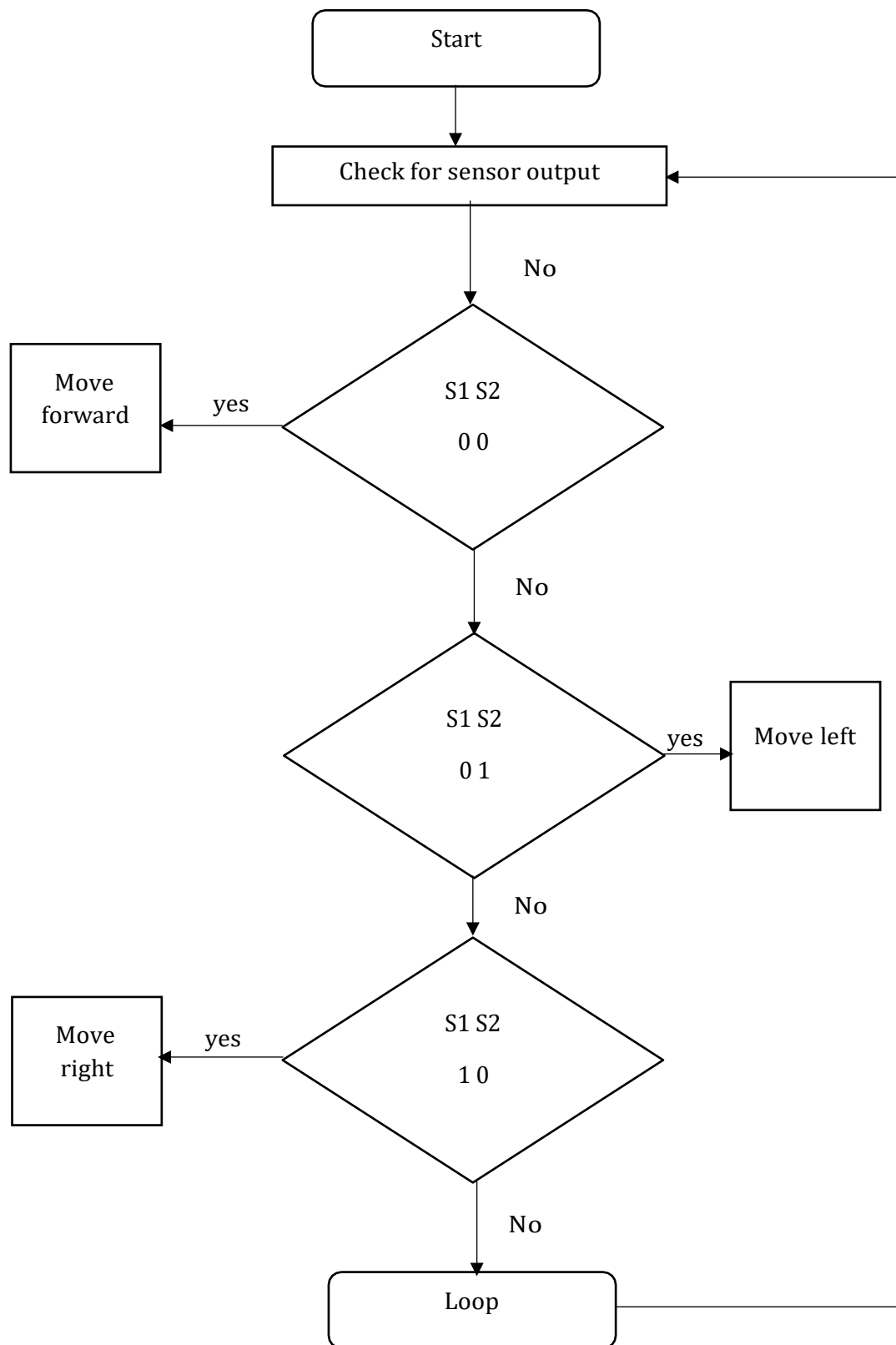
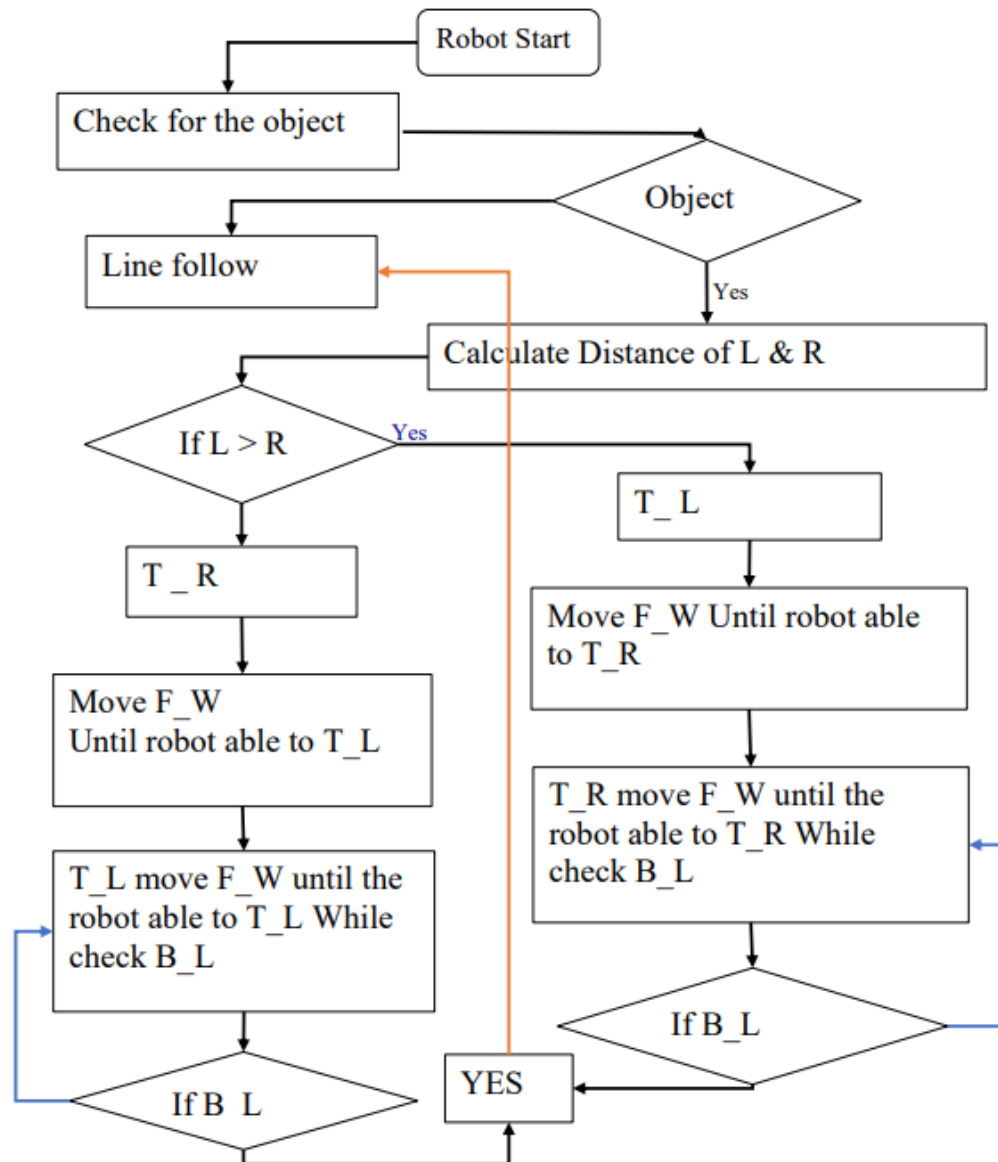


Fig 2: System design and Block diagram

7.2 Flowchart of Line follower :



7.3 Flow Chart of Obstacle Detection



R = Right.

L = Left.

T_L = Turn Left.

T_R = Turn Right.

F_W = Forward Move.

B_L = Black Line

Implementation

8.1 Hardware Implementation

In this work we implemented two types of robots, First we describe implementation of the obstacle avoidance robot, then describe the implementation of the line following robot. The differences between these two robots are what task the robot solves and what sensor the robot uses to solve the task. The objective of obstacle avoidance robot is to avoid the collisions with different objects. The robot uses a proximity sensor to measure the distance to the objects. The line following robot main goal is to stay on the path which is the black tape on the floor. This robot uses the IR sensor to see the difference between the light floor and the dark tape placed on the floor.

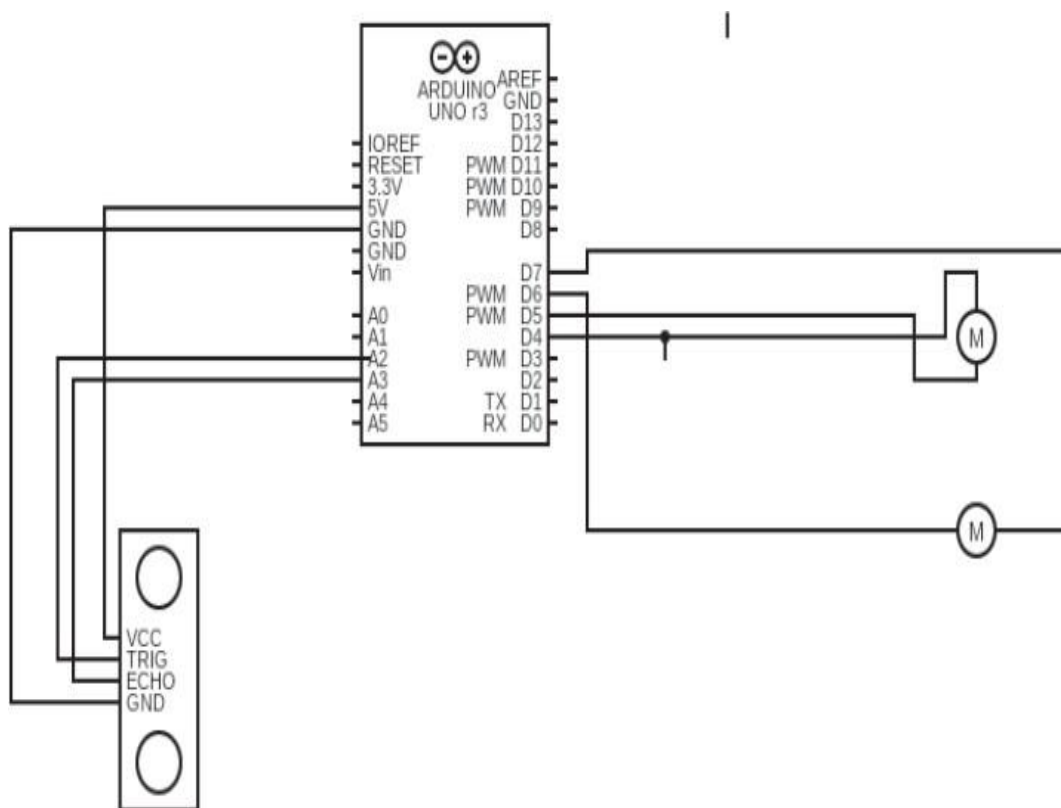


Figure : Circuit Diagram of Obstacle Avoidance Robot.

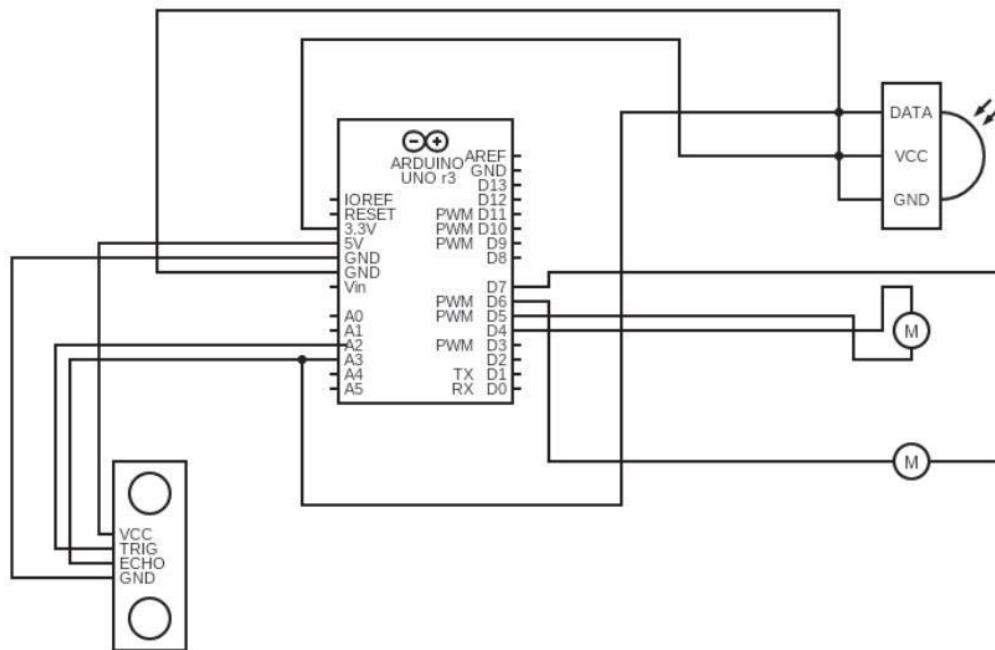


Figure : Circuit Diagram of Line following robot

8.2 Software Implementation

In this software implementation, we have used C and C++. Using this code, we have designed our line following robot.

Code:

```
#define enA 10//Enable1 L298 Pin enA
#define in1 9 //Motor1 L298 Pin in1
#define in2 8 //Motor1 L298 Pin in1
#define in3 7 //Motor2 L298 Pin in1
#define in4 6 //Motor2 L298 Pin in1
#define enB 5 //Enable2 L298 Pin enB

#define L_S 4 //ir sensor Left
#define R_S 3 //ir sensor Right
#define echo A0 //Echo pin
#define trigger A1 //Trigger pin

#define servo A2

int Set=25;
int distance_L, distance_F, distance_R;

void setup(){ // put your setup code here, to run once

Serial.begin(9600); // start serial communication at 9600bps

pinMode(R_S, INPUT); // declare if sensor as input

Obstacle Avoiding Line Follower Car
```

```

pinMode(L_S, INPUT); // declare ir sensor as input

pinMode(echo, INPUT );// declare ultrasonic sensor Echo pin as input
pinMode(trigger, OUTPUT); // declare ultrasonic sensor Trigger pin as Output

pinMode(enA, OUTPUT); // declare as output for L298 Pin enA
pinMode(in1, OUTPUT); // declare as output for L298 Pin in1
pinMode(in2, OUTPUT); // declare as output for L298 Pin in2
pinMode(in3, OUTPUT); // declare as output for L298 Pin in3
pinMode(in4, OUTPUT); // declare as output for L298 Pin in4
pinMode(enB, OUTPUT); // declare as output for L298 Pin enB

analogWrite(enA, 110); // Write The Duty Cycle 0 to 255 Enable Pin A for Motor1 Speed
analogWrite(enB, 110); // Write The Duty Cycle 0 to 255 Enable Pin B for Motor2 Speed

pinMode(servo, OUTPUT);

for (int angle = 70; angle <= 140; angle += 5) {
    servoPulse(servo, angle); }
for (int angle = 140; angle >= 0; angle -= 5) {
    servoPulse(servo, angle); }

for (int angle = 0; angle <= 70; angle += 5) {
    servoPulse(servo, angle); }

distance_F = Ultrasonic_read();

delay(500);
}

void loop(){
//=====
//   Line Follower and Obstacle Avoiding
//=====

distance_F = Ultrasonic_read();
Serial.print("D F=");Serial.println(distance_F);

//if Right Sensor and Left Sensor are at White color then it will call forward function
if((digitalRead(R_S) == 0)&&(digitalRead(L_S) == 0)){
    if(distance_F > Set){forward();}
    else{Check_side();}
}

//if Right Sensor is Black and Left Sensor is White then it will call turn Right function
else if((digitalRead(R_S) == 1)&&(digitalRead(L_S) == 0)){turnRight();}

//if Right Sensor is White and Left Sensor is Black then it will call turn Left function
else if((digitalRead(R_S) == 0)&&(digitalRead(L_S) == 1)){turnLeft();}

delay(10);
}

void servoPulse (int pin, int angle){
int pwm = (angle*11) + 500;    // Convert angle to microseconds
digitalWrite(pin, HIGH);
delayMicroseconds(pwm);
digitalWrite(pin, LOW);
delay(50); // Refresh cycle of servo
}

```

```

*****Ultrasonic_read*****
long Ultrasonic_read(){
  digitalWrite(trigger, LOW);
  delayMicroseconds(2);
  digitalWrite(trigger, HIGH);
  delayMicroseconds(10);
  long time = pulseIn (echo, HIGH);
  return time / 29 / 2;
}

void compareDistance(){
  if(distance_L > distance_R){
    turnLeft();
    delay(500);
    forword();
    delay(600);
    turnRight();
    delay(500);
    forword();
    delay(600);
    turnRight();
    delay(500);
  }
  else{
    turnRight();
    delay(500);
    forword();
    delay(600);
    turnLeft();
    delay(500);
    forword();
    delay(600);
    turnLeft();
    delay(500);
  }
}

void Check_side(){
  Stop();
  delay(100);
  for (int angle = 70; angle <= 140; angle += 5) {
    servoPulse(servo, angle); }
    delay(300);
    distance_R = Ultrasonic_read();
    Serial.print("D R=");Serial.println(distance_R);
    delay(100);
  for (int angle = 140; angle >= 0; angle -= 5) {
    servoPulse(servo, angle); }
    delay(500);
    distance_L = Ultrasonic_read();
    Serial.print("D L=");Serial.println(distance_L);
    delay(100);
  for (int angle = 0; angle <= 70; angle += 5) {
    servoPulse(servo, angle); }
    delay(300);
    compareDistance();
  }
}

void forword(){ //forword
digitalWrite(in1, LOW); //Left Motor backward Pin
digitalWrite(in2, HIGH); //Left Motor forward Pin
  Obstacle Avoiding Line Follower Car

```

```
digitalWrite(in3, HIGH); //Right Motor forward Pin
digitalWrite(in4, LOW); //Right Motor backward Pin
}
```

```
void backward(){ //backward
digitalWrite(in1, HIGH); //Left Motor backward Pin
digitalWrite(in2, LOW); //Left Motor forward Pin
digitalWrite(in3, LOW); //Right Motor forward Pin
digitalWrite(in4, HIGH); //Right Motor backward Pin
}
```

```
void turnRight(){ //turnRight
digitalWrite(in1, LOW); //Left Motor backward Pin
digitalWrite(in2, HIGH); //Left Motor forward Pin
digitalWrite(in3, LOW); //Right Motor forward Pin
digitalWrite(in4, HIGH); //Right Motor backward Pin
}
```

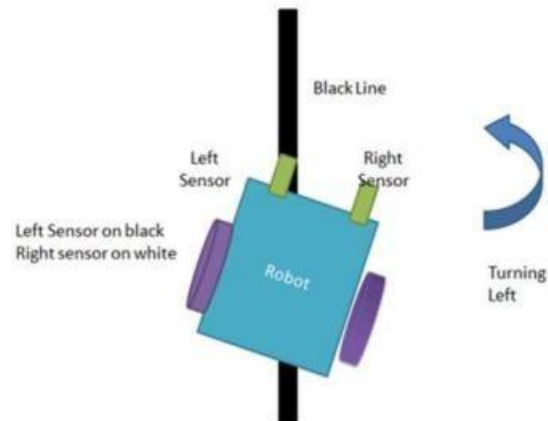
```
void turnLeft(){ //turnLeft
digitalWrite(in1, HIGH); //Left Motor backward Pin
digitalWrite(in2, LOW); //Left Motor forward Pin
digitalWrite(in3, HIGH); //Right Motor forward Pin
digitalWrite(in4, LOW); //Right Motor backward Pin
}
```

```
void Stop(){ //stop
digitalWrite(in1, LOW); //Left Motor backward Pin
digitalWrite(in2, LOW); //Left Motor forward Pin
digitalWrite(in3, LOW); //Right Motor forward Pin
digitalWrite(in4, LOW); //Right Motor backward Pin
}
```

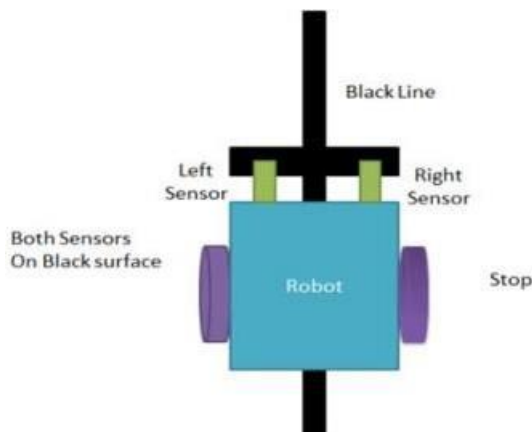
8.3 Working of Robot

Now, how the robot run on black line and avoid the obstacle in front of it.

Step 1 - Right motor stops while left motor continues to travel when left sensor enters white (for black line tracer) zone, causing the robot to turn to the right and return to the white line show in Figure.

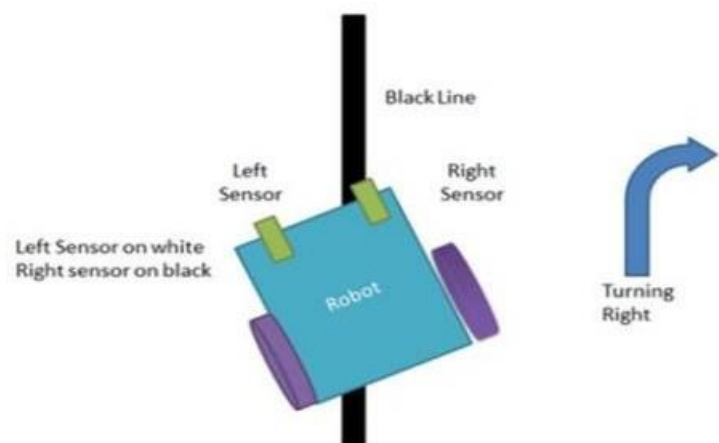


Step 2 - Since the first sensor on the right is facing the black line, its response will be low, while that of the other sensors will be high. i.e., the left wheel is allowed to move freely while the right wheel is kept stationary until the centre sensor's reaction drops.



Step 3 - When the right sensor enters the white area, the left motor stops while the right motor keeps moving, causing the robot to turn to the left and return to the white line show in Figure left.

Step 4 - The middle sensor will always be on the line, and as the line is black in color, it won't reflect the radiation back, thus its response will always be low. In contrast, the responses of the other two sensors, which will be on the bright surface, will always be high. Robot advances when both sensors are on the black line show in Figure right.



Step 5 - When none of the three lines are identified, the robot moves in a circle until one is located if all three sensors are on a brighter surface in which case, they will all be high.

Step 6 - To examine the impediment from three aspects, the robot head rotates 180 degrees. This robot's brain is an Arduino board that uses information from an ultrasonic sensor to determine how far an obstacle is from the robot. Arduino determines the direction of the obstacle based on the servo motor's angle.

Step 7 - Measuring the distances in both directions, it chooses the path that passes through the obstruction with the biggest gap. It advances in that direction, off the path, and travels around an obstruction until it crosses a black line. It finds the black line and continues along its path. When the distance between the object and the robot is closer than the threshold distance, which is set in the programmer, the obstacle-avoiding module is activated.

CONCLUSION

- The project successfully designed and implemented an obstacle-avoiding line follower car using Arduino.
- The car can accurately follow a black line on the ground while detecting and avoiding obstacles.
- The combination of IR sensors for line following and an ultrasonic sensor for obstacle detection ensures efficient navigation.

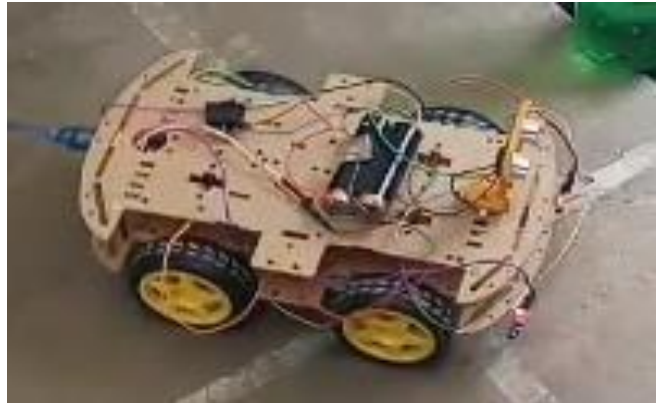


Fig 1 : Structure of Line follower Car



Fig 2 : Car following the path



Fig 3 : Car detecting the obstacle



Fig 4 : Car overcoming the obstacle

ADVANTAGES

- **Automation:** The robot autonomously follows the line and avoids obstacles, reducing the need for manual control.
- **Efficiency:** It efficiently navigates predefined paths, making it useful for warehouse automation or guided transportation.
- **Versatility:** The same platform can be extended for other applications (e.g., home security, surveillance).

LIMITATIONS

- **Line Intensity:** The robot relies on a sufficiently visible black line; low-intensity lines may affect performance.
- **Sensor Damage:** Malfunctioning IR sensors or damaged components can impact accuracy.
- **Mechanical Range:** The robot can only follow angled lines within its mechanical constraints.

APPLICATION

1. Warehouse Automation:

- In warehouses or factories, these robots can transport goods along predefined paths without human intervention.
- They follow lines on the floor, avoiding obstacles (such as other robots or pallets) to optimize material handling.

2. Surveillance and Security:

- Deploying these robots in large facilities (e.g., museums, shopping malls, or parking lots) allows continuous monitoring.
- They can follow specific routes, detect intruders, and report suspicious activities.

3. Hazardous Environments:

- In environments with toxic gases, radiation, or extreme temperatures, these robots can navigate safely.
- They follow paths while avoiding hazardous areas or obstacles.

4. Unmanned Vehicle Navigation:

- These principles extend to autonomous vehicles (cars, drones, or ground robots).
- Line following and obstacle avoidance are essential for self-driving cars and delivery drones.

5. Educational Platforms:

- Schools and universities use similar projects to teach robotics, programming, and sensor integration.
- Students learn about sensors, motor control, and algorithm design.

6. Entertainment and Competitions:

- Line-following robot races or competitions challenge participants to build efficient and accurate robots.
- These events encourage creativity and problem-solving skills.

FUTURE SCOPE

- **Advanced Algorithms:** Implement more sophisticated algorithms for obstacle avoidance, such as SLAM (Simultaneous Localization and Mapping).
- **Camera Integration:** Enhance obstacle detection using cameras or depth sensors.
- **Wireless Communication:** Add communication modules (Bluetooth, Wi-Fi) for remote control or data exchange.
- **Machine Learning:** Train the robot to adapt to different environments and optimize its behaviour.

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- Line Follower Robots – Controlling, Working Principle and Applications <https://www.elprocus.com/line-follower-robot-basics-controlling/>

- **Books :**
 - i) Arduino Book for Beginners by Mike Cheich
 - ii) Arduino made Simple with Interactive Projects by Ashwin Pajankar
 - iii) Build A Line Follower Robot : A User-Friendly Guide Publisher: Southern University College Editor: CHIA KIM SENG, KANG ENG SIEW, ENG PEI CHEE, HUI SENG KEONG

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 - iv. <https://www.arduino.cc/reference/en/>
 - v. <https://projecthub.arduino.cc/lightthedreams/line-following-robot-34b1d3>